

CASE REPORT

Fatal diphenidol poisoning: a case report and a retrospective study of 16 cases

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Abstract Diphenidol hydrochloride (DPN), a nonphenothiazinic antiemetic agent used primarily in patients with Meniere disease and labyrinthopathies to treat vomiting and vertigo, is considered to be a relatively safe drug. Since it was first approved in the United States in 1967, this drug has been widely used in Latin America and Asia and has contributed to sporadic suicidal and accidental poisonings in mainland China and Taiwan. However, its toxic or lethal concentration ranges have not yet been determined. We report a case of a 23-year-old female who suffered from DPN poisoning that resulted in death. At autopsy, there were no typical pathological findings, except for cerebral edema with high acetylcholinesterase expression. Postmortem analysis of DPN revealed 45 µg/ml in heart blood, 39 µg/ml in femoral vein blood, 141 µg/g in the liver, and 53 mg in the gastric contents. These concentrations indicated that the cause of death was DPN poisoning. The circumstances indicated that the manner of death was suicide. We also present a retrospective study, in which we review and summarize the literature from 1998 to 2014 and describe 16 cases of poisoning, including information from autopsy reports and postmortem drug concentrations. In

forensic practice, drug residues at the scene, patients with convulsions and disturbance of consciousness, and rapidly occurring deaths, should draw attention to the possibility of this drug. Toxicological analysis and the exclusion of other diseases may ultimately be used to confirm DPN poisoning.

Keywords Diphenidol hydrochloride · Poison · Death · Acetylcholinesterase

Introduction

Diphenidol hydrochloride (DPN) [1, 1-diphenyl-4-piperidino-1-butanol hydrochloride] (Fig. 1), commonly called diphenidol (molecular formula $C_{21}H_{27}NO \cdot HCl$) and marketed under the brand names Cephadol and Vontrol, is an anti-emetic agent that was first approved for use in the United States in 1967 [1]. It has been widely used in Latin American and Asian markets for the treatment of vertigo associated with inner-ear disturbances such as labyrinthitis, vestibular neuronitis and Meniere disease [2, 3]. DPN can also diminish vestibular stimulation, depress labyrinthine function, and suppress nausea and vomiting at the medullary chemoreceptive trigger zone [4]. Moreover, improvement of blood flow in vertebral arteries has also been noted.

DPN is an odorless, sour, white crystalline powder that easily dissolves in methanol, ethanol, and thermal water. The most common routes of exposure to DPN are oral, rectal, and parenteral administration [5]. The chemical properties of DPN tablets are stable, and it is rapidly metabolized. The therapeutic dose is 25–50 mg, administered no more than three times per day. The peak plasma DPN concentration usually occurs between 1.5 and 3 h after oral administration, and it takes 2 h to reach a peak

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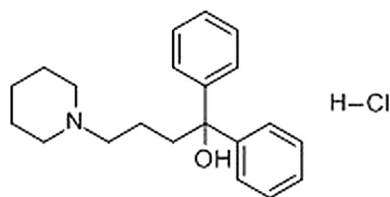


Fig. 1 Molecular structure of diphenidol hydrochloride

plasma concentration of 0.35 µg/ml after a single dose of 100 mg DPN. Its elimination half-life is approximately 4 h [6]. No information on its toxic and lethal ranges in humans has been previously reported. Nevertheless, fatal consequences of DPN overdose due to suicidal or accidental reasons have been sporadically reported in mainland China and Taiwan. Thus, clinical studies and medico-legal research on the use and toxicity of DPN continue to be important to the diagnosis and treatment of DPN poisoning.

Case report

History

A 23-year-old female was found convulsing in a hotel room 1 h after lunch and was sent to hospital. During lunch, she drank a glass of rice wine and complained of dizziness. Physical examinations revealed mydriasis, and her heart rate and pulse were undetectable. Nikethamide and lidocaine were employed in an emergency treatment. The woman was pronounced dead 1 h later. Investigation of the scene revealed 5 empty strips of DPN tablets (each strip contained 24 tablets, and each tablet contained 25 mg DPN, with a total of 120 tablets equal to 3000 mg) in her room.

Autopsy

Medico-legal autopsy was performed at 14 h postmortem. Bilateral mydriasis was found. Cyanosis of the lips and nails was obvious. The brain was moderately edematous. Congestion of the lungs, spleen, kidneys, and other organs was observed.

Histopathology

Moderate cerebral edema, mild neuron necrosis, neuronophagia, and astrocytic hyperplasia were observed. Vascular fibrous necrosis, perivascular hemosiderin, and amyloid corpuscles of the subventricular zone were obvious. Moderate edema was found in the white matter of the lentiform nucleus (Fig. 2a). Immunohistochemistry showed that acetylcholinesterase (AChE) was highly expressed (Fig. 2b). Pulmonary edema and

emphysema were evident. Contraction band necrosis of the heart was observed. Her liver cells had fatty changes. Congestion and micro-emboli were found in multiple organs.

Toxicology

Whole blood samples from the heart and femoral veins, as well as liver and gastric contents, were collected during autopsy and maintained at 4 °C prior to and after analysis. Her bladder was empty; thus, a urine sample could not be acquired.

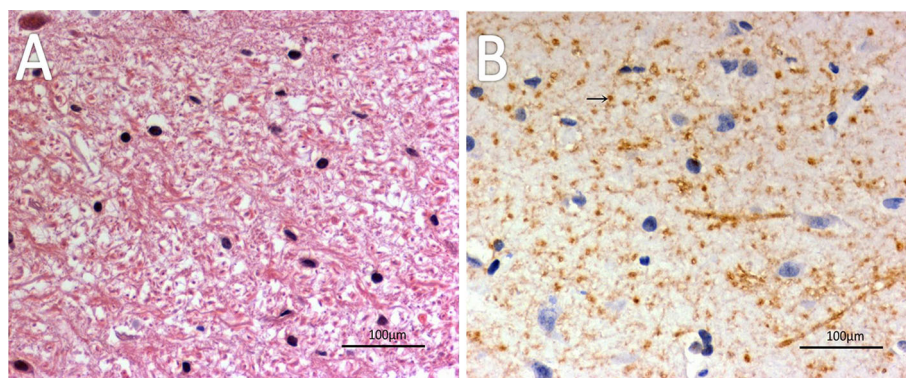
A postmortem blood sample was screened for a selection of benzodiazepines, barbiturates, organophosphates, and tetramine by gas chromatography-mass spectrometry (GC–MS). Screening analysis for blood ethanol was performed using a head-space gas chromatograph equipped with a flame ionization detector (HSGC–FID). Information from the case indicated that the deceased had ingested many DPN tablets. After being alkalinized, the blood sample was also analyzed by GC–MS for a selection of alkaline drugs.

Determination of DPN concentration

DPN (purity >99.6 %) and SKF_{525A} (internal standard, purity >98 %) were supplied by the National Institutes for Food and Drug Control (Beijing, China). Methanol, ethyl acetate and *n*-hexane were obtained from Aladdin® (Los Angeles, USA). Deionized water was obtained using a Milli-Q water purification system (Millipore, Darmstadt, Germany). All other chemicals were of analytical grade and were purified by distillation. Control human whole blood for preparation of the standard samples was supplied by the Wuhan Blood Centre (Wuhan, China), and a control liver sample was provided by the Department of Forensic Medicine of Tongji Medical College (Wuhan, China). Stock solutions of DPN and SKF_{525A} were prepared in methanol. Working standards were prepared in water containing 0.9 % NaCl. Six calibration samples were prepared from whole blood spiked with working standard solutions (10.0–100.0 µg/ml).

To extract DPN, 50 µl of an internal standard solution containing SKF_{525A} (100 µg/ml) and 50 µl NaOH (10 mol/l) were added to 500 µl whole blood in a centrifuge tube. After being alkalinized, the aqueous phase was extracted with 4 ml of a mixture of *n*-hexane:ethyl acetate (7:3, v/v). After the mixture was vortexed for 2 min and centrifuged at 3000 rpm for 5 min, the upper organic layer was evaporated to dryness under a slow stream of nitrogen at 40 °C. The dried extracts were reconstituted in 50 µl methanol. The reconstituted tissue extracts were centrifuged at 3000 rpm for 5 min. A 1 µl aliquot was injected into the GC–NPD system.

Fig. 2 Cerebral edema (lentiform nucleus) was moderate (**a** HE, $\times 400$), with high expression of AChE (**b** IHC, $\times 400$)



The samples were analyzed with an Agilent 7820 series gas chromatograph, equipped with an autosampler (HP7683) and an NPD system. High-purity nitrogen was used as the carrier gas at a constant flow of 4 ml/min. A DB-5 5 % phenyl-methyl siloxane capillary column (30 m $\times$ 0.32 mm I.D. and 0.25 μ m film thickness) was used. The initial oven temperature was 120 $^{\circ}$ C for 1 min. Then, the temperature was programmed to increase to 280 $^{\circ}$ C at 15 $^{\circ}$ C/min, after which it was held for 5 min (overall run time of 17 min). The temperatures of the injector and detector were maintained at 280 and 300 $^{\circ}$ C, respectively. The flow rates were 4.0 ml/min for nitrogen gas, 60 ml/min for air, and 3.0 ml/min for hydrogen. Quantification was performed with Agilent Open LAB using GC Chemstation C.01.05 software. The retention times were 10.10 and 9.62 min for DPN and SKF_{525A}, respectively. The standard curve for DPN spiked into human whole blood was linear over a concentration range of 10.0–100.0 μ g/ml. The equation was $y = 0.0506x - 0.0824$, and the average correlation coefficient (r) was 0.998. The detection limit and limit of quantitation, based on a signal-to-noise ratio of $S/N = 3$ and $S/N = 10$, were 1.0 and 10.0 μ g/ml, respectively. Intra-day and inter-day precision was within 5 % (the samples were analyzed over 3 days).

Toxicological findings

Routine toxicological analysis revealed the presence of nikethamide, lidocaine and ethanol (0.127 mg/ml) in the postmortem whole blood (Fig. 3). No other compounds included in the standard analytical program were detected.

DPN was found in the heart blood (45 μ g/ml), femoral vein blood (39 μ g/ml), liver (141 μ g/g), and gastric contents (53 mg) after a more comprehensive analysis.

Forensic investigation indicated that this woman was in a marriage crisis and had revealed suicidal intentions once previously. According to investigation of the scene, she had ingested at least 120 DPN tablets, and toxicological

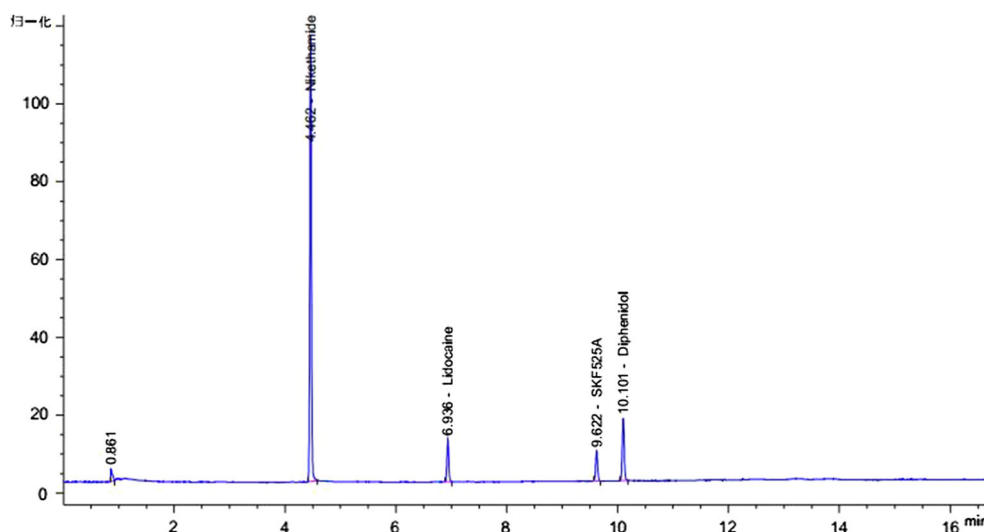
analysis revealed no other narcotic drugs in her body. DPN is not typically chosen as a homicidal drug because of its sour taste. Thus, we excluded the possibility of homicidal or accidental administration. The cause of death was declared to be intoxication with DPN, and it was considered to be a case of suicidal poisoning.

Discussion

DPN is an effective drug used for the treatment of vertigo and dizziness. This drug is commercially available at both prescription and non-prescription doses. We searched PubMed, Web of Science, Medline, and China National Knowledge Infrastructure for all years and all languages using the keywords “Diphenidol” and “poison.” We found 16 necropsy or quantitative reports of cases of DPN poisoning that were published from 1998 to 2014 [7–15] (Table 1). All of the cases were written in Chinese except for one, which was written in English. These cases involved 8 males and 9 females aged 2.5–29 years (median of 21 years). According to these reports, 7/17 occurred by suicide (all females with a median age of 24 years) and 8/17 were accidental (all children with a median age of 3.4 years). In our review, all of the poisoning incidents involving children occurred accidentally, but they were deliberate in the females with suicidal intentions. The easy access to DPN and potential risk of poisoning constitute major concerns in mainland China and Taiwan in contrast with other parts of the world, where these issues have not been given a substantial amount of attention. Appropriate controls should be established based on the general characteristics of the clinical data, pathological findings, and toxicokinetics.

In forensic practice, DPN poisoning can be difficult to diagnose. DPN is commonly used because of its pharmacological function and safety, but mortality associated with its use has not yet been examined in the literature. Clinical manifestations of DPN poisoning include initial inhibition

Fig. 3 GC-NPD chromatogram of a blood sample spiked with SKF525A



of the central nervous system (CNS), presenting as sedation or somnolence, before the key symptoms of convulsions and unconsciousness begin. This latent period can last for approximately 2 h and tends to result in fatality due to a delay in treatment. Additionally, clinical symptoms and signs of DPN poisoning vary and are easily confused with epilepsy, tetanus, rabies, hysteria, and rodenticide poisoning. These 16 cases have revealed that DPN fatality occurs unexpectedly within a very short period of time, but most autopsies have reported negative findings. DPN has not been included in routine toxicological screening because its poisoning is poorly understood in the health care sector in Western countries. If there are no tablets, drug strips, or bottles left on the scene, a definitive diagnosis of DPN poisoning may be overlooked.

Studies have reported that DPN preparations have H1 receptor antagonistic, anticholinergic, and quinidine-like effects [16–18]. Serious adverse effects caused by an elevated dose include hallucination and confusion, and drowsiness, dry mouth, depression, restlessness, headache, and transitory hypotension also occasionally occurs [7]. The precise mechanism underlying these effects remains unclear. However, a tri-phasic pattern termed “inhibition-excitation-inhibition” triggered by administration of a large dose may promote successive H1 receptor antagonist and anticholinergic effects in the CNS that result in consciousness disturbance, convulsions and respiratory failure. The H1 receptor antagonistic effect of DPN causes inhibition of the CNS and relaxation of the vascular smooth muscle, leading to a sedated or somnolent state. An overdose can induce arrhythmia, including QT interval prolongation, T wave change, U wave appearance, atrioventricular block, bundle branch block, ventricular premature contraction, ventricular tachycardia and ventricular fibrillation. Due to the similar configuration of

DPN as atropine, strong peripheral anticholinergic activity resulting from an overdose also causes hypotension, arrhythmia, and convulsions. Additionally, DPN may block synaptic transmission to keep a high level of cholinesterase in the CNS [19]. Lastly, with an increase in the concentration, respiratory depression is worsened, ultimately leading to death due to respiratory and circulatory failure. Therefore, common postmortem findings of DPN poisoning include cyanosis of the face and nails, organ congestion, cerebral and pulmonary edema, and degeneration of the liver and renal tubular epithelial cells. The peripheral anticholinergic activity leads to mydriasis and dry lips and throat. The quinidine-like toxic effects may cause myocardial necrosis.

Because the quantitative measurement of the DPN concentration was generally promoted in 2005 [6], 6/17 studies tested the postmortem drug concentrations in heart blood (6/6 and mean of 17.17 µg/ml), urine (1/6 and mean of 3.60 µg/ml), liver (2/6 and mean of 130.35 µg/g), and gastric contents (1/6 and mean of 53 mg). In our case, high concentrations of DPN were detected in the heart blood, femoral vein blood, liver, and gastric contents. DPN intoxication was assumed to be the primary cause of death. DPN toxicity in humans needs further characterization, and no toxic or lethal ranges have been defined. The concentrations in our case were 45 µg/ml in the heart blood and 141 µg/g in the liver; thus, they were higher than those that had been previously reported. In addition, ethanol was positively detected. Specific interactions between DPN and ethanol are unknown. However, aggravated poisoning might have occurred because ethanol is a good solvent of DPN that increases the absorption rate through the gastrointestinal tract. This increased absorption may have resulted in the high concentration in the blood and low concentration in the gastric contents that were observed.

Table 1 Reported cases of difenidol hydrochloride poison (1998–2015)

Authors and publication year	Age/sex	Cause	Exposure route	Primary symptoms	Autopsy and pathology findings	Actual dosage	Postmortem sampled time	Postmortem drug concentration
Yang et al. (1998)	2.5/M	Accident	Digestive system	Recurrent seizures, hypotension, respiratory failure and coma	None	225 mg	None	None
Jiang et al. (2002)	3–29/ M:F = 6:2	Suicide 1/8; accident 7/8	Digestive system	Convulsion 8/8, conscious disturbance 8/8, sinus tachycardia 8/8, mydriasis 7/8, dyspnea 6/8, facial and lips cyanosis 6/8, blood pressure drop 4/8, restlessness 4/8	Facial and lips cyanosis, mydriasis, congestion of organs, severe edema of lungs and brain, degeneration of liver cell and renal tubular epithelial cell	Suicide 2625 mg; accident 445–650 mg	Not mentioned	Not mentioned
Zhang et al. (2006)	20/F	Not mentioned	Digestive system	Convulsion, unconsciousness	Edema and congestion of organs, myocardial necrosis	Not mentioned	Not mentioned	Not mentioned
Xiong et al. (2007)	25/F	Suicide	Digestive system	Clonic convulsion, heart rate fast; slow breathing rhythm	Facial and limbs cyanosis	3000 mg	5 h	Not mentioned
Liu (2008)	22/F	Suicide	Digestive system	Clonic convulsion, coma, facial cyanosis	Edema of organs, limbs cyanosis	6000 mg	7 days	Heart blood 14.04 µg/ml, liver 25.69 µg/g
Liu et al. (2009)	Not mentioned	Suicide	Digestive system	Not mentioned	Edema and congestion of organs, facial, lips and limbs cyanosis, vomiting, frothy fluid in trachea and lungs, petechia in lungs	1250 mg	Not mentioned	Heart blood 19.07 µg/ml
Wang et al. (2010)	23/F	Suicide	Digestive system	Clonic convulsion, vomiting	Cerebral vascular congestion, disappearance of Nissl body, neuronophagia, satellitism, pulmonary vascular congestion, edema and hemorrhage, myocardial necrosis	Not mentioned	11 h	Heart blood 13.3 µg/ml, urine 3.6 µg/ml, gastric contents 725 µg/g, liver 235 µg/g
Zhang et al. (2011)	25/F	Not mentioned	Digestive system	Not mentioned	Not mentioned.	>2500 mg	Not mentioned	Heart blood 8.56 µg/ml

Table 1 continued

Authors and publication year	Age/sex	Cause	Exposure route	Primary symptoms	Autopsy and pathology findings	Actual dosage	Postmortem sampled time	Postmortem drug concentration
Zheng et al. (2014)	24/F	Suicide	Digestive system	Not mentioned	Edema and congestion of organs, petechia in heart and lungs	>774 mg	Not mentioned	Heart blood 2.71 µg/ml
Current case	23/F	Suicide	Digestive system	Convulsion	Edema, micro-embolus and congestion of organs, myocardial necrosis, cerebral angiospasm	3000 mg	13 h	Heart blood 45 µg/ml, femoral vein blood 39 µg/ml, liver 141 µg/g, gastric contents 53 mg

We also speculated that the differences in the postmortem concentrations in heart blood among studies might be attributable to individual variations, including differences in weight and the physical properties of gastric contents in the presence of the drug, the use of co-medication, and the intervals between DPN administration and death and between death and sample collection. Additional studies are required to confirm these possibilities.

DPN is an alkaline and fat-soluble drug that is rapidly absorbed but is distributed and eliminated slowly; thus, it is prone to redistribution. Oral administration is the primary poisoning route; therefore, the gastric concentration is typically relatively higher than the concentration in any other organ. In our case study, autopsy was performed 13 h after death. The anatomic distance from the heart to the stomach is shorter than that from the stomach to the femoral vein, making postmortem dissemination easier. Thus, the concentration in the heart blood was higher than that in the femoral blood.

Wang et al. [20] have reported that the plasma DPN concentrations for therapeutic use (50 mg by oral administration) are 0.32 ± 0.16 µg/ml (single dose) and 0.36 ± 0.18 µg/l (multiple doses) in the peripheral blood. The lethal concentration ranges in our case and retrospective studies ranged from 2.71 to 45 µg/ml (17.11 ± 14.74 µg/ml, 774–6000 mg dosage) in the heart blood. The lethal concentration is much higher than the therapeutic concentration. We speculate that the toxic concentration might be in this range, but confirmed values have not yet been reported.

In conclusion, DPN poisoning should be considered in patients with convulsions and consciousness disturbance as their key symptoms, and who die rapidly. The history of toxin exposure, clinical features, and presence of toxins should be analyzed to confirm poisoning. In a patient with DPN poisoning, electrocardiogram typically reveals arrhythmia and paroxysmal ventricular tachycardia. An abnormal electroencephalogram, identified by θ band amplification, is beneficial to dynamically observe the cerebral injury, illness fluctuation, therapeutic effect, and prognosis. The differential diagnosis includes CNS diseases and other types of poisoning. Blood, urine, liver, and gastric content specimens should be obtained for toxicological analysis by high-performance liquid chromatography (HPLC) or GC–MS [6, 21]. DPN is unaffected by temperature and antimicrobial agents but decomposes spontaneously. A total of 90 % of DPN is expelled through urine in its original state [22]; thus, the antemortem collection of blood and urine during therapy is appropriate.

Key points

1. Diphenidol is a relatively safe drug but can cause death in a suicidal or accidental manner.
2. Acetylcholinesterase is highly expressed in the white matter of the lentiform nucleus.

3. Scene investigations, clinical symptoms, death process, and toxicological analysis are performed to confirm diphenidol poisoning.

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